

Cat of CSC 1263 Discrete Structures for Computing

1st Year Computer a Engineering. Max: 50pts Duration: 2h. Answer all questions

Question 1. 10pts

- A. The sum of the first 3 terms of an arithmetic series is 21 and the sum of the next three terms is 66. [4pts]
- a) Find the value of the first term and the common difference.
 - b) Write an expression for the n^{th} term of the series u_n .
 - c) Find the sum of the first 10 terms.
- B. Let $P(n)$ be the statement that use **mathematical induction** to show that $1 + 2 + \dots + n = n(n + 1)/2$ for the positive integer n . [3pts]
- a) What is the statement $P(1)$?
 - b) Show that $P(1)$ is true, completing the basis step of the proof.
 - c) What is the inductive hypothesis?
 - d) What do you need to prove in the inductive step?
 - e) Complete the inductive step, identifying where you use the inductive hypothesis.
- C. Find a formula for the n th term of the following geometric sequence. What is the ninth term of the sequence? 5, 15, 45, ... [3pts]

Question 2. 10pts

A. Find the solutions of:

a. $S = \sum_{j=3}^6 (2 + 4j)$ [3pts] b. $S = \sum_{j=3}^5 \sum_{i=1}^4 (4j - i)$ [3pts]

B. Let p , q , and r be the propositions

p : You have the flu.

q : You miss the final examination

r : You pass the course

Express each of these propositions as an English sentence.

$(p \wedge q) \vee (\neg q \wedge r)$ [2pts]

- C. Find 5 sequence number of pseudorandom numbers generated by the linear congruential method with modulus $m = 9$, multiplier $a = 7$, increment $c = 4$, and seed $x_0 = 3$. [2pts]

Question 3. 10pts

- a) Use the Euclidean algorithm to calculate $\gcd(100, 270)$. Show each step. [2pts]
- b) Show that $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ is a tautology [2pts]
- c) The sum of the first three terms of an arithmetic series is 21 and the sum of the next three terms is 66. What is the twentieth term? [3pts]
- d) b. Show that if n is positive integer then $C(n, 3) = P(n, 2)$ [3pts]

Question 4. 10pts

- a) Encrypt the plaintext message "STOP GLOBAL WARMING" using the shift cipher with shift $k = 11$. [3pts]

- b) Decrypt the ciphertext message “LEWLYPLUJL PZ HNYLHA ALHJOLY” that was encrypted with the shift cipher with shift $k = 7$ [3pts]
- c) Find the solutions of the simultaneous system of recurrence relations $a_n = a_{n-1} + b_{n-1}$ $b_n = a_{n-1} - b_{n-1}$ with $a_0 = 1$ and $b_0 = 2$. [4pts]

Question 5. 10pts

- a. Prove that the sum of the terms of the arithmetic progression: [5pts]

$$S = \sum_{j=1}^n (a + jd) = na + d \frac{n(n+1)}{2}$$

- b. Let f and g be the functions from the set of integers to the set of integers defined by: $f(x) = 2x + 3$ and $g(x) = 3x^2 + 2$. What is the composition of f and g^{-1} ? What is the composition of g and f^{-1} ? [2pts]
- c. In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class? [3pts]

Good Luck!!!!